Course Website

114-2

自然語言處理與應用

https://github.com/mcps5601/CGUNLP_2026_Spring

自然語言處理與應用
Natural Language Processing and Applications

Course Introduction

Instructor: 林英嘉 (Ying-Jia Lin)
2026/02/23

Slido
NLP0223 (Link)

About Me

Experience and Education (Time)	
Assistant Professor	Department of Artificial Intelligence, Chang Gung University (2025/02 -)
Postdoctoral Researcher	Department of Computer Science, National Tsing Hua University (2024/09 - 2025/01)
Ph.D.	Department of Computer Science and Information Engineering, National Cheng Kung University (2019/08 - 2024/02)
M.S.	Institute of Biomedical Informatics, National Yang-Ming University (2017/09 - 2019/07)
B.S.	Department of Biomedical Sciences, Chang Gung University (2013/09 - 2017/06)

Outline

- Introduction to Natural Language Processing
- Introduction to Large Language Models
- Course Introduction and requirements

什麼是自然語言處理（Natural Language Processing, NLP）？

文字搜尋

下一個字預測

程式語言 vs. 自然語言

程式語言

main transformers / src / transformers / loss / loss_deformable_detr.py

Code Blame 178 lines (151 loc) · 7.15 KB

```python

from ..image_transforms import center_to_corners_format
from ..utils import is_scipy_available
from .loss_for_object_detection import (
    HungarianMatcher,
    ImageLoss,
    _set_aux_loss,
    generalized_box_iou,
    sigmoid_focal_loss,
)


if is_scipy_available():
    from scipy.optimize import linear_sum_assignment


class DeformableDetrHungarianMatcher(HungarianMatcher):
    @torch.no_grad()
    def forward(self, outputs, targets):
        """
        Differences:
        - out_prob = outputs["logits"].flatten(0, 1).sigmoid() instead of softmax
```

自然語言

先生不知何許人也，亦不詳其姓字。宅邊有五柳樹，因以為號焉

　　閑靜少言，不慕榮利。好讀書，不求甚解，每有會意，便欣然忘食。性嗜酒，家貧，不能常得。親舊知其如此，或置酒而招之。造飲輒盡，期在必醉，既醉而退，曾不吝情去留。環堵蕭然，不蔽風日；短褐穿結，簞瓢屢空。——晏如也。常著文章自娛，頗示己志。忘懷得失，以此自終。

　　贊曰：黔婁之妻有言：「不戚戚於貧賤，不汲汲於富貴。」極其言，茲若人儔乎？酣觴賦詩，以樂其志。無懷氏之民歟！葛天氏之民歟！

什麼是自然語言處理？

天青色等煙雨

I like apple!

?

定義 ➡️

自然語言處理（Natural Language Processing, NLP）是讓電腦能「理解、分析、產生」人類語言的技術。

The Revolution of ChatGPT

ChatGPT came out in November, 2022.

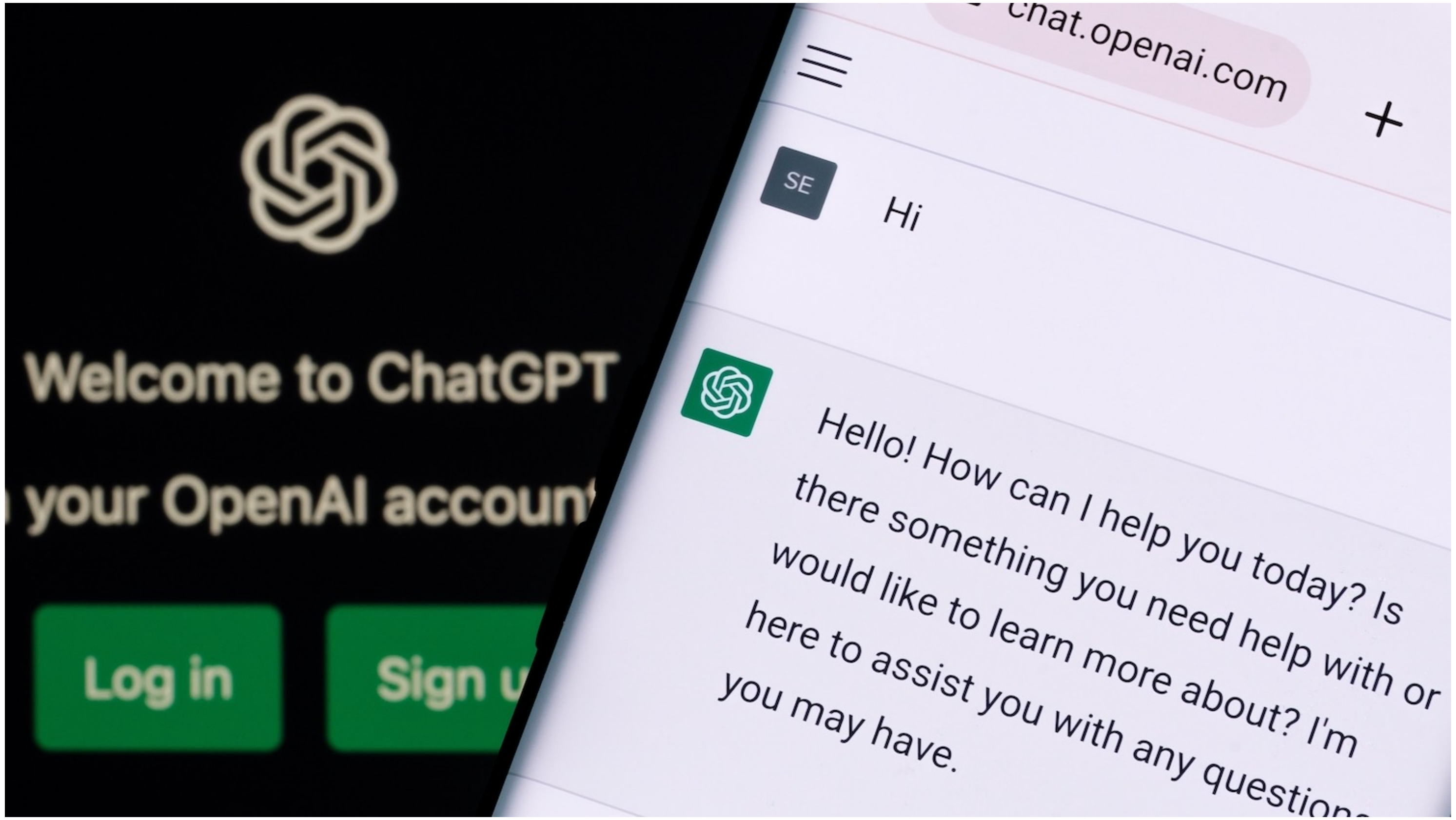
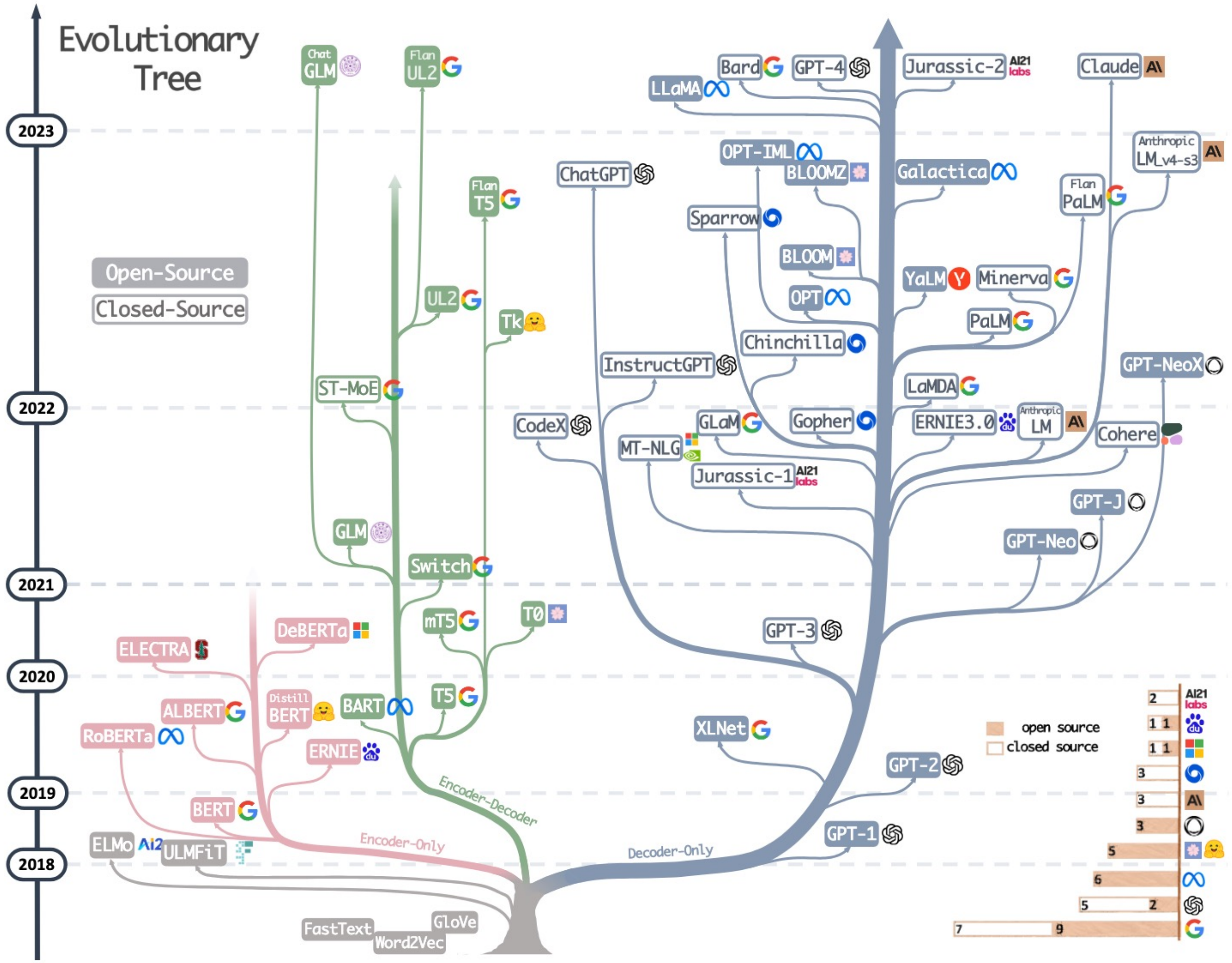


Figure source: https://poole.ncsu.edu/thought-leadership/article/lets-chat-about-chatgpt/
https://openai.com/index/chatgpt/

大型語言模型的演化樹



Yang, Jingfeng, et al. "Harnessing the power of llms in practice: A survey on chatgpt and beyond." ACM Transactions on Knowledge Discovery from Data 18.6 (2024): 1-32.

NLP

Chinese LLMs

SuperCLUE：2024年值得关注的中文大模型全景图

Figure source: https://chinai.substack.com/p/chinai-264-superclue-april-2024-report

SuperCLUE Leaderboard

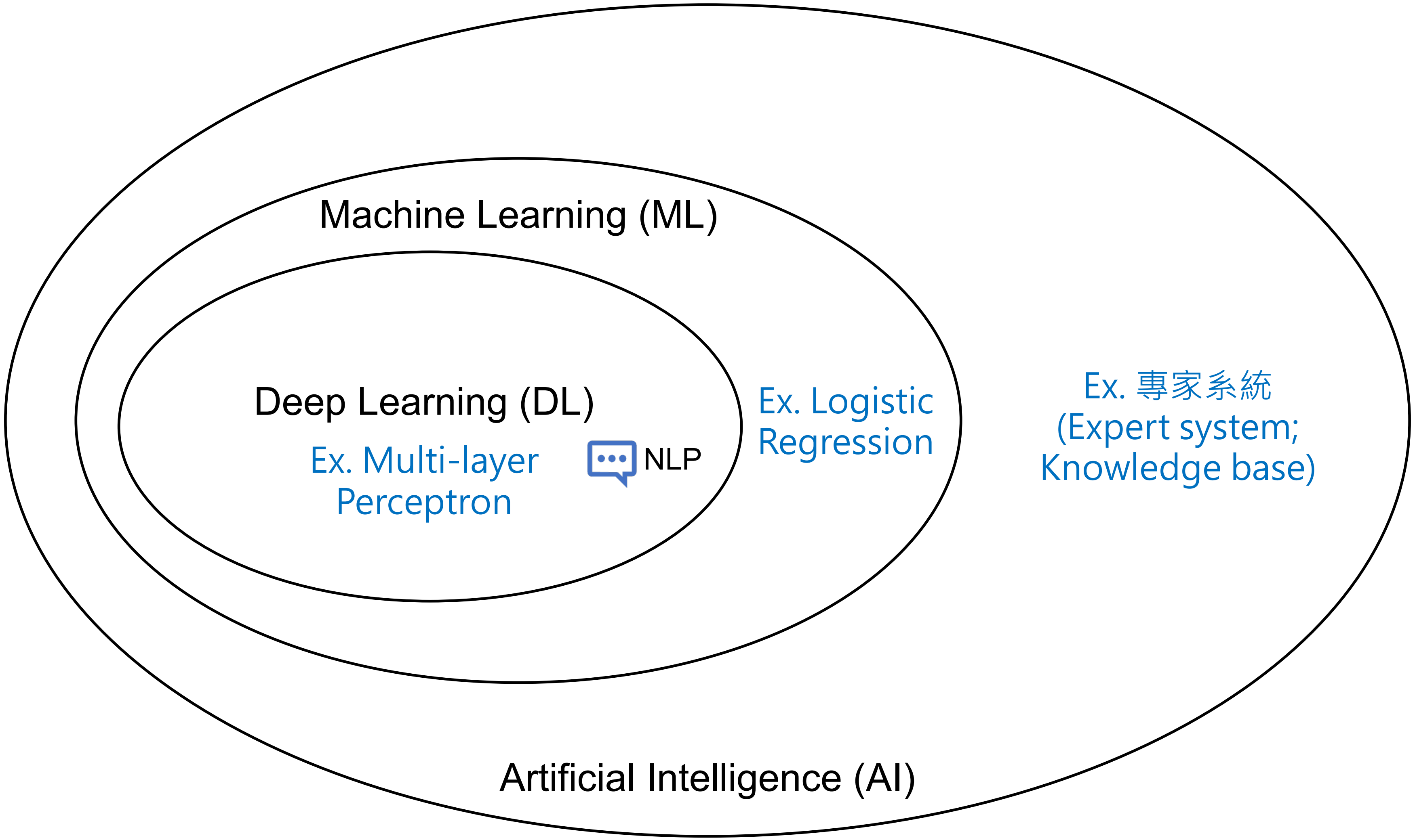
为减少波动影响，榜单将分差1分值内的模型视为并列排名。国外模型与部分国内模型不参与排名，只做参考。

排名	模型名称	机构	开/闭源	总分	数学推理	幻觉控制	科学推理	精确指令遵循	代码生成	智能体(任务规划)	属地	使用方式	是否推理	发布日期
-	Claude-Opus-4.6(high effort)	Anthropic	闭源	71.93	77.88	85.87	78.63	51.78	59.91	77.52	海外	API	是	2026.2.11
-	Claude-Opus-4.5-Reasoning	Anthropic	闭源	68.25	73.04	88.31	73.77	51.10	48.40	74.87	海外	API	是	2026.1.28
-	Gemini-3-Pro-Preview	Google	闭源	65.59	80.87	83.16	73.77	43.56	47.17	65.02	海外	API	是	2026.1.28
-	GPT-5.2(high)	OpenAI	闭源	64.32	72.04	88.56	75.21	37.81	30.91	81.39	海外	API	是	2026.1.28
1	Kimi-K2.5-Thinking	月之暗面	开源	61.50	77.39	78.54	67.21	24.45	53.33	68.06	国内	API	是	2026.1.29
-	Gemini-3-Flash-Preview	Google	闭源	61.43	79.13	81.57	74.17	39.45	40.64	53.59	海外	API	是	2026.1.28
1	Qwen3-Max-Thinking	阿里巴巴	闭源	60.61	80.87	74.05	68.85	28.22	41.56	70.13	国内	API	是	2026.1.29
1	GLM-5	智谱AI	开源	60.50	73.91	81.05	67.21	27.35	51.35	62.12	国内	API	是	2026.2.12
2	Doubao-Seed-1.8-251228(Thinking)	字节跳动	闭源	58.17	68.70	80.37	68.85	32.60	40.33	58.15	国内	API	是	2026.1.28
2	DeepSeek-V3.2-Thinking	深度求索	开源	57.55	72.17	76.57	71.31	29.86	39.84	55.53	国内	API	是	2026.1.28
3	GLM-4.7	智谱AI	开源	56.22	68.42	83.85	66.39	21.37	41.26	56.03	国内	API	是	2026.1.28
3	ERNIE-5.0	百度	闭源	56.18	65.22	74.61	64.75	37.53	45.63	49.32	国内	API	是	2026.1.28
-	Grok-4	X.AI	闭源	55.23	74.78	78.90	68.03	14.79	49.51	45.36	海外	API	是	2026.1.28
4	Kimi-K2-Thinking	月之暗面	开源	54.02	70.43	69.39	62.30	16.99	46.00	59.01	国内	API	是	2026.1.28
5	Qwen3-Max-Preview-Thinking	阿里巴巴	闭源	52.46	61.74	62.01	63.93	24.66	37.99	64.40	国内	API	是	2026.1.28
-	Grok-4.1-Fast(Reasoning)	X.AI	闭源	51.59	71.30	71.08	62.30	16.44	34.36	54.05	海外	API	是	2026.1.28
6	Tencent HY 2.0 Think	腾讯	闭源	50.22	65.79	70.91	66.39	18.90	37.32	41.99	国内	API	是	2026.1.29
6	LongCat-Flash-Thinking-2601	美团	开源	49.22	62.62	66.76	60.66	15.07	37.81	52.43	国内	API	是	2026.1.28
7	Step-3.5-Flash	阶跃星辰	开源	48.97	75.65	57.04	63.93	10.68	36.64	49.88	国内	API	是	2026.2.5
7	Qwen3-Max-2025-09-23	阿里巴巴	闭源	48.46	62.61	64.58	61.48	6.30	47.23	48.59	国内	API	否	2026.1.28

Figure source: https://www.superclueai.com/

自然語言處理（廣義）屬於一機器學習領域的任務

Venn Diagram about AI, ML, and DL



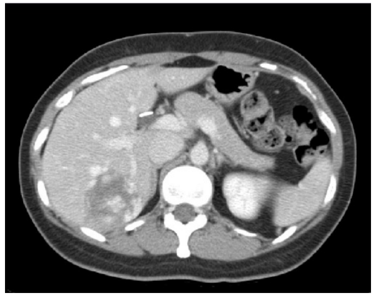
What is Deep Learning?

- We start the concepts from machine learning.

Machine Learning function!

Classifier

Cat ☑
Dog
Appl
e



Classifier

Malignant
Benign ☑

How about DL in Natural Language Processing?

Sentiment Analysis (IMDb dataset [1])

This movie was awful. The ending was absolutely horrible. → Classifier → Positive / Negative ☑

Translation, EN-ZH (WMT-19 dataset [2])

Yet not nearly enough has been invested in this effort. → Machine Translation Model → 但目前這方面的投入還遠遠不夠。

[1] https://huggingface.co/datasets/stanfordnlp/imdb
[2] https://huggingface.co/datasets/wmt/wmt19/viewer/zh-en

Why do we need DL in NLP?

79	**RNTN**	85.4	×	Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank	2013
80	**SWEM-concat**	84.3	×	Baseline Needs More Love: On Simple Word-Embedding-Based Models and Associated Pooling Mechanisms	2018
81	**MV-RNN**	82.9	×	Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank	2013
82	**GloVe+Emo2Vec**	82.3	×	Emo2Vec: Learning Generalized Emotion Representation by Multi-task Training	2018
83	**Emo2Vec**	81.2	×	Emo2Vec: Learning Generalized Emotion Representation by Multi-task Training	2018
84	**ToWE-CBOW**	78.8	×	Task-oriented Word Embedding for Text Classification	2018
85	**Joined Model Multi-tasking**	54.72	×	Exploring Joint Neural Model for Sentence Level Discourse Parsing and Sentiment Analysis	2017

https://paperswithcode.com/sota/sentiment-analysis-on-sst-2-binary
Liu, Qian, et al. "Task-oriented word embedding for text classification." Proceedings of the 27th international conference on computational linguistics. 2018.

電腦是如何學習自然語言呢？

影像 vs. 文字

⟶ 28 x 28 的矩陣

I want to study AI ⟶ ?

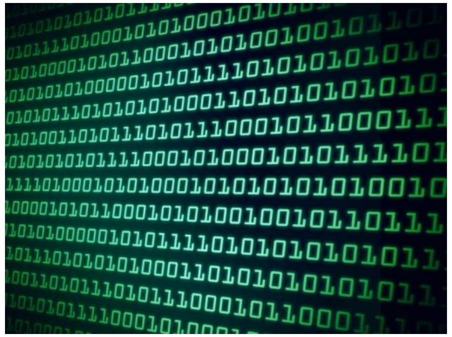


Figure source: https://kopu.chat/wp-content/uploads/2017/04/binary-507x380.jpeg

Word Representations

like

apple

?

Basic Approach

Corpus

- I like to eat apple
- You like to watch TV

Vocabulary

apple	[1, 0, 0, 0, 0, 0, 0, 0]
eat	[0, 1, 0, 0, 0, 0, 0, 0]
I	[0, 0, 1, 0, 0, 0, 0, 0]
like	[0, 0, 0, 1, 0, 0, 0, 0]
watch	[0, 0, 0, 0, 1, 0, 0, 0]
to	[0, 0, 0, 0, 0, 1, 0, 0]
⋮	

One hot encodings with each vector size equal to the vocab size (8 in this case)

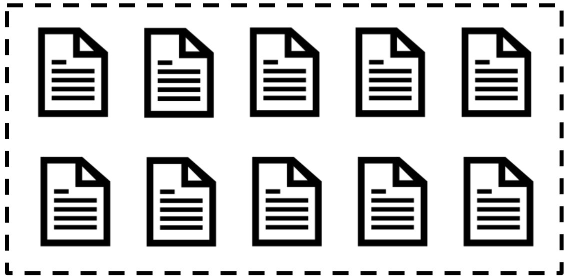
Pros: handy, training-free

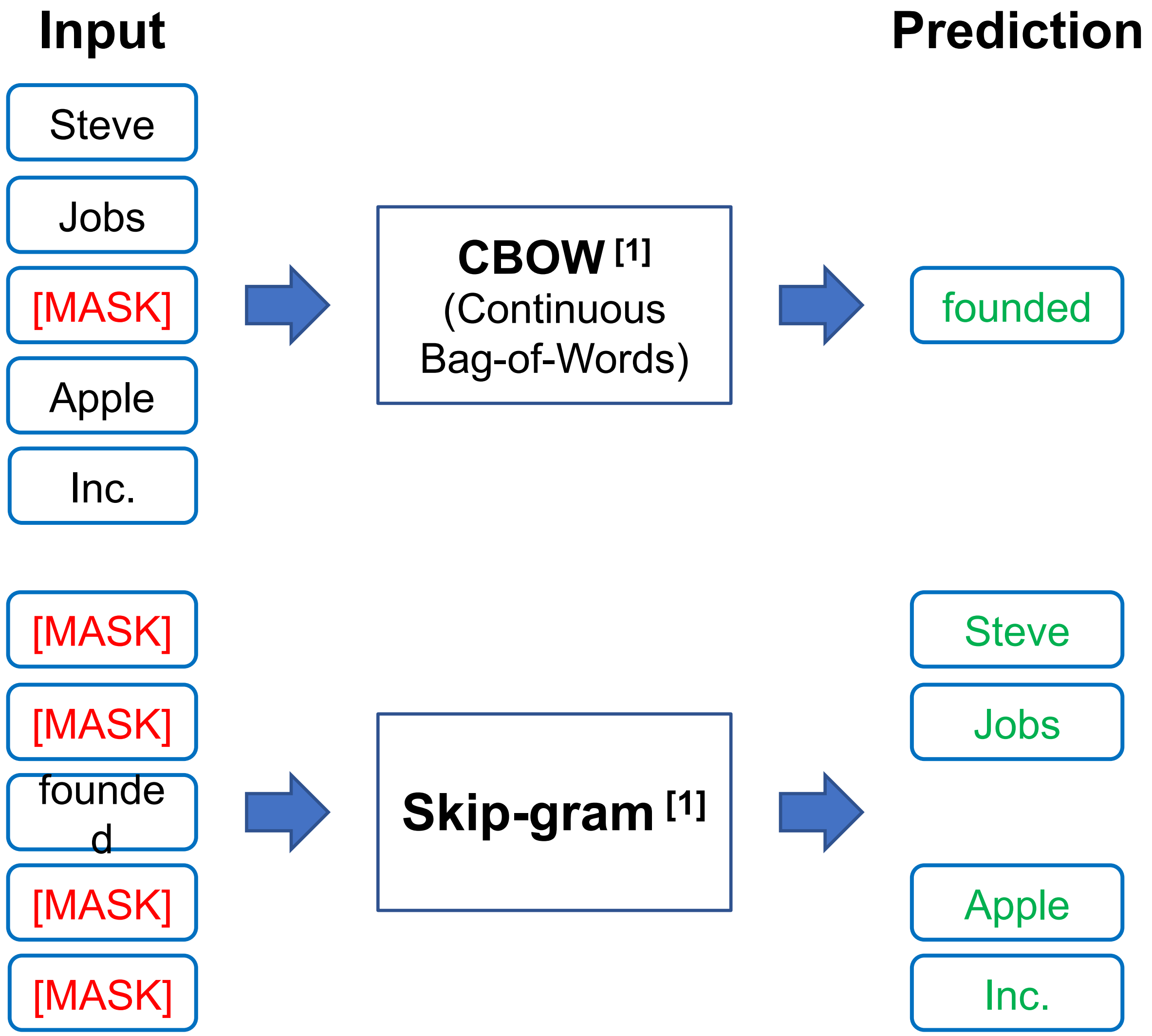
Cons: sparse matrix which occupies much memory

詞嵌入：把詞嵌入到向量 (Word2Vec)

[1] Mikolov et al. "Distributed representations of words and phrases and their compositionality." NeurIPS 2013.

非常大量的語料庫
(例如：Wikipedia)





每個字，都是一段向量 – 詞向量

- Word2Vec 的範例長相:

Dimension size (e.g., 300)

Vocabulary size (e.g., 30,000)

apple -0.110960 0.016115 -0.004809 0.033589 0.121455 …
banana -0.027713 -0.015676 0.003314 0.077602 0.159718 …
…
…
…

Word Embeddings 視覺化

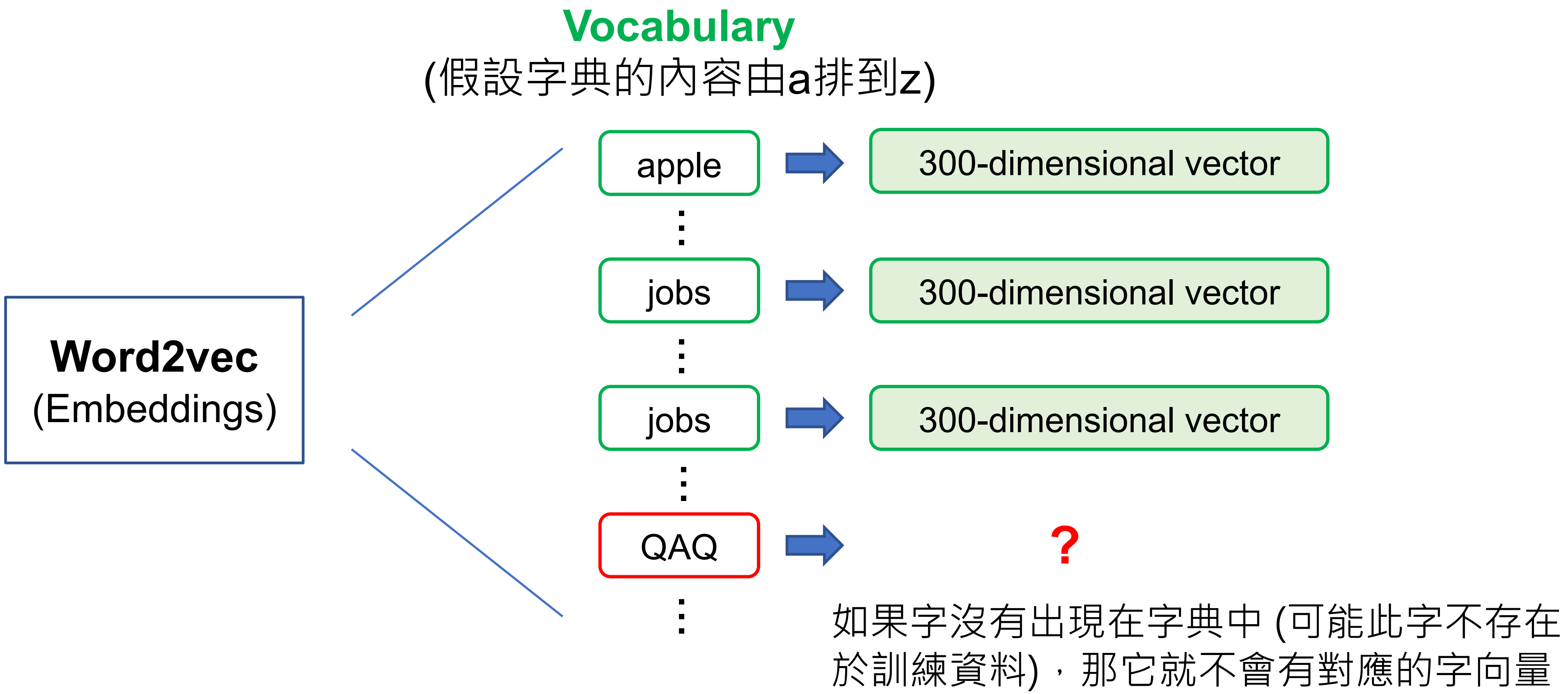
- **Word embedding model**: glove-wiki-gigaword-100
- **Dimension reduction**: t-SNE
- **Dataset**: Mikolov et al., 2013

Problems of Traditional Word Embeddings

1. Out-of-vocabulary problem
2. Model architecture for down-stream applications
   - Context-dependent representations

The Out-of-vocabulary Problem

- Before training word embeddings, words are split by whitespaces for English corpora



Sub-word Tokenization (Byte Pair Encoding)

- Byte Pair Encoding (BPE)[3] is a sub-word (子詞) tokenization technique used to handle rare words by breaking them into more frequent sub-word units.

Example sentence: I printed Hello world.

Traditional word tokenization

I	printed	Hello	world

Traditional with BPE

I	prin	ted	Hell	o	world

分詞方法的決定是無監督式的

[3] Sennrich et al. "Neural Machine Translation of Rare Words with Subword Units." ACL 2016.

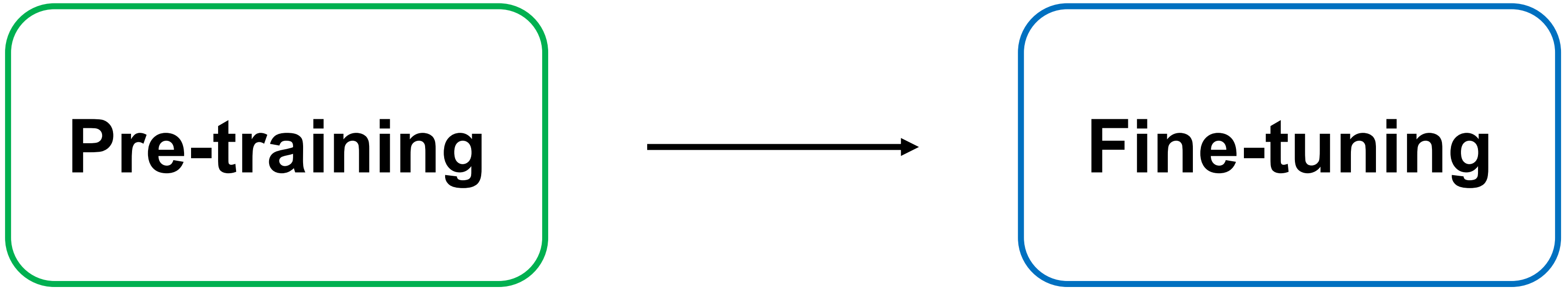
Sub-word Tokenization 的好處

- With sub-word tokenization algorithms like BPE, we can handle representations for unknown words (or mis-spelled words).
- In machine translation , the compound word issues between source and target languages can be alleviated.
- State-of-the-art pre-trained language models (e.g., GPT-3, BERT) adopt sub-word tokenization algorithms before pre-training.

Problems of Traditional Word Embeddings

1. Out-of-vocabulary problem
2. Model architecture for down-stream applications
   - Context-dependent representations

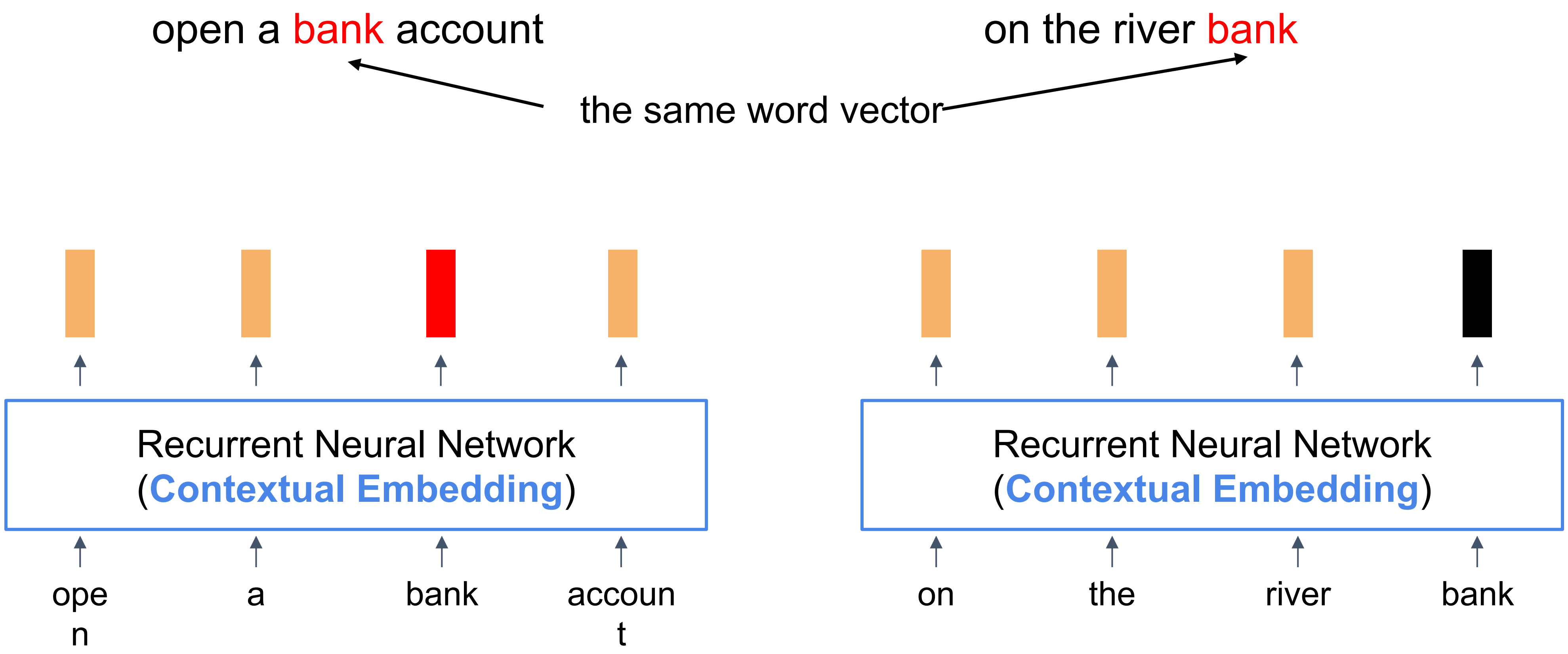
The Pre-training then Fine-tuning Paradigm



在大量資料上進行訓練，通常是自監督式 (Self-Supervised Training)

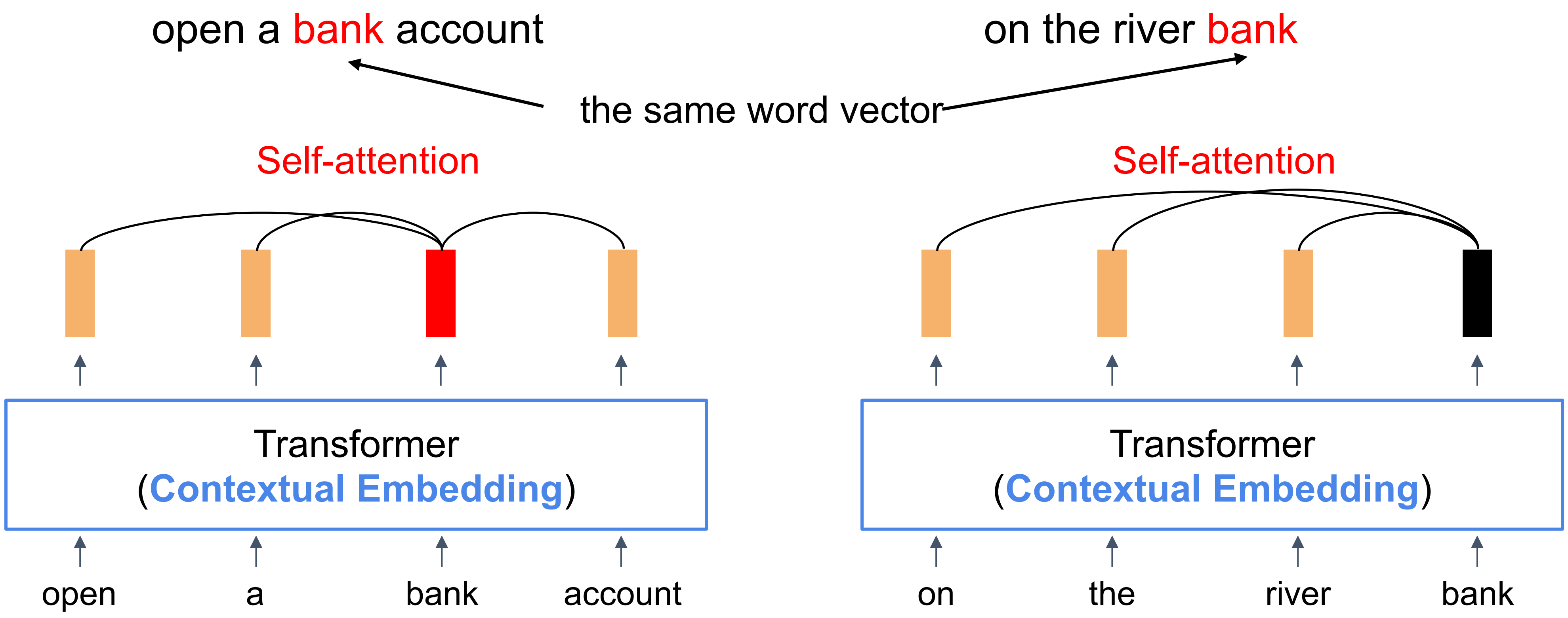
在目標資料上進行訓練，通常是監督式 (Supervised Training)，也就是需要有標註的資料才能進行模型訓練

Contextual Word Representations (RNN)



How about bi-directional?

Contextual Word Representations (Transformer)



Model Training: Pre-training and Fine-Tuning

Step1: Pre-training (use large-scale corpora)

Open a [MASK] account → Model (e.g., Word2vec, BERT) → bank

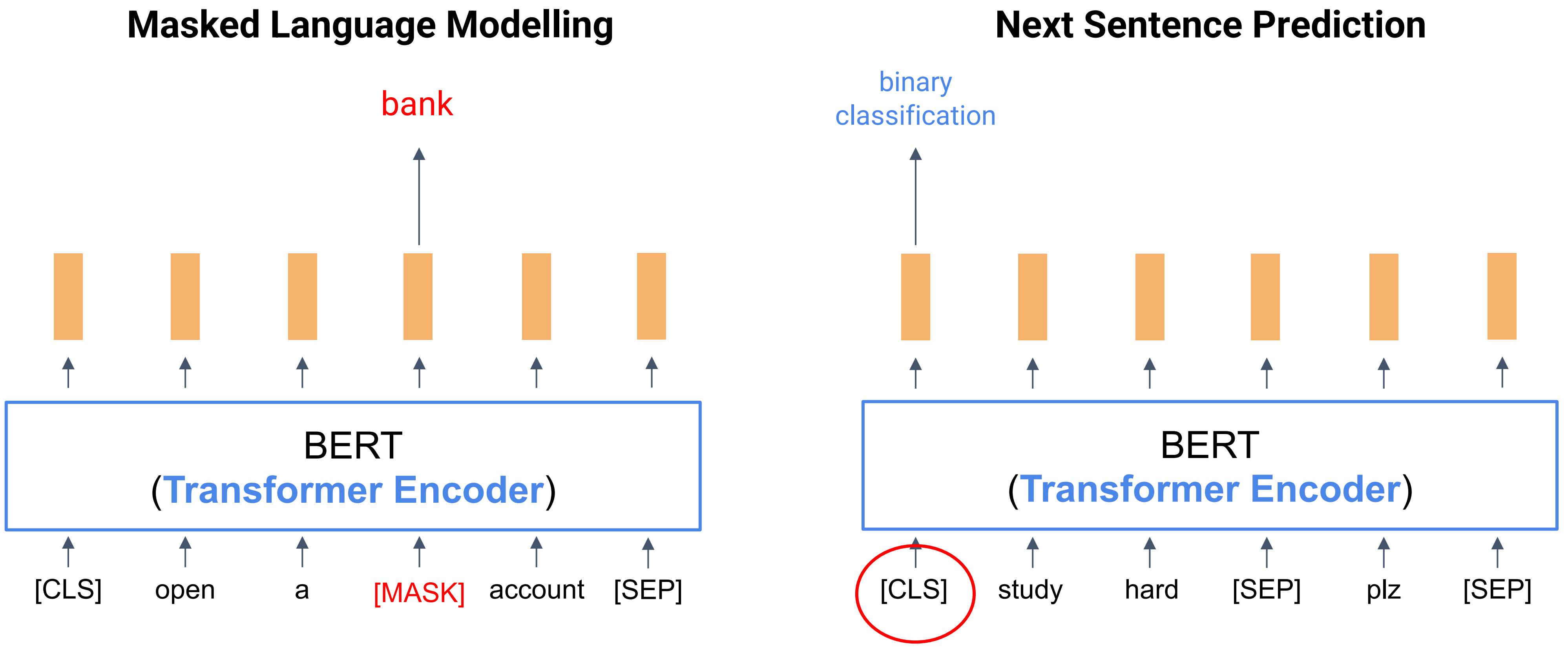
Step2: Fine-Tuning (use datasets from target tasks)

It was a good movie! → Model (e.g., Word2vec + RNN; BERT + Classifier) → Positive

Devlin, Jacob, et al. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." NAACL 2019.

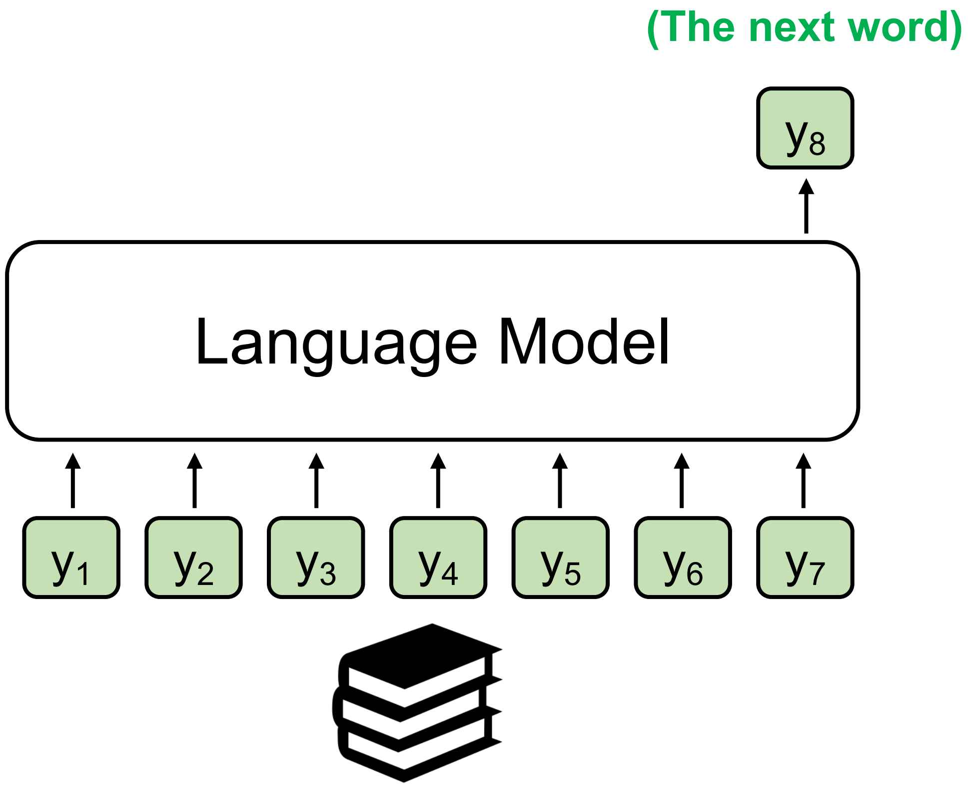
BERT (Bidirectional Encoder Representations from Transformers)

- BERT 有兩種預訓練任務：



Devlin, Jacob, et al. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." NAACL 2019.

Generative Pre-Training (GPT)



$$P(y_t | y_1, y_2, \dots, y_{t-1})$$

- A model that assigns probabilities to upcoming words is called **a language model**.
- The task involving predictions of upcoming words is **language modeling**.

GPT 系列作品（有透露細節的）時間線

2018/06/11	(GPT-1) Improving language understanding by generative pre-training
2019/02/14	(GPT-2) Language models are multi-task learners
2020/05/28	(GPT-3) Language models are few-shot learners
2022/05/04	(InstructGPT) Training language models to follow instructions with human feedback

以前的 GPT 有口吃

InstructGPT: ChatGPT 的前身

Prompt

Explain the moon landing to a 6 year old in a few sentences.

Completion

GPT-3

```
Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.
```

InstructGPT

```
People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.
```

Figure source: https://openai.com/index/instruction-following/
Ouyang, Long, et al. “Training language models to follow instructions with human feedback.” NeurIPS 2022.

The Pre-training then Fine-tuning Paradigm

Fine-tuning

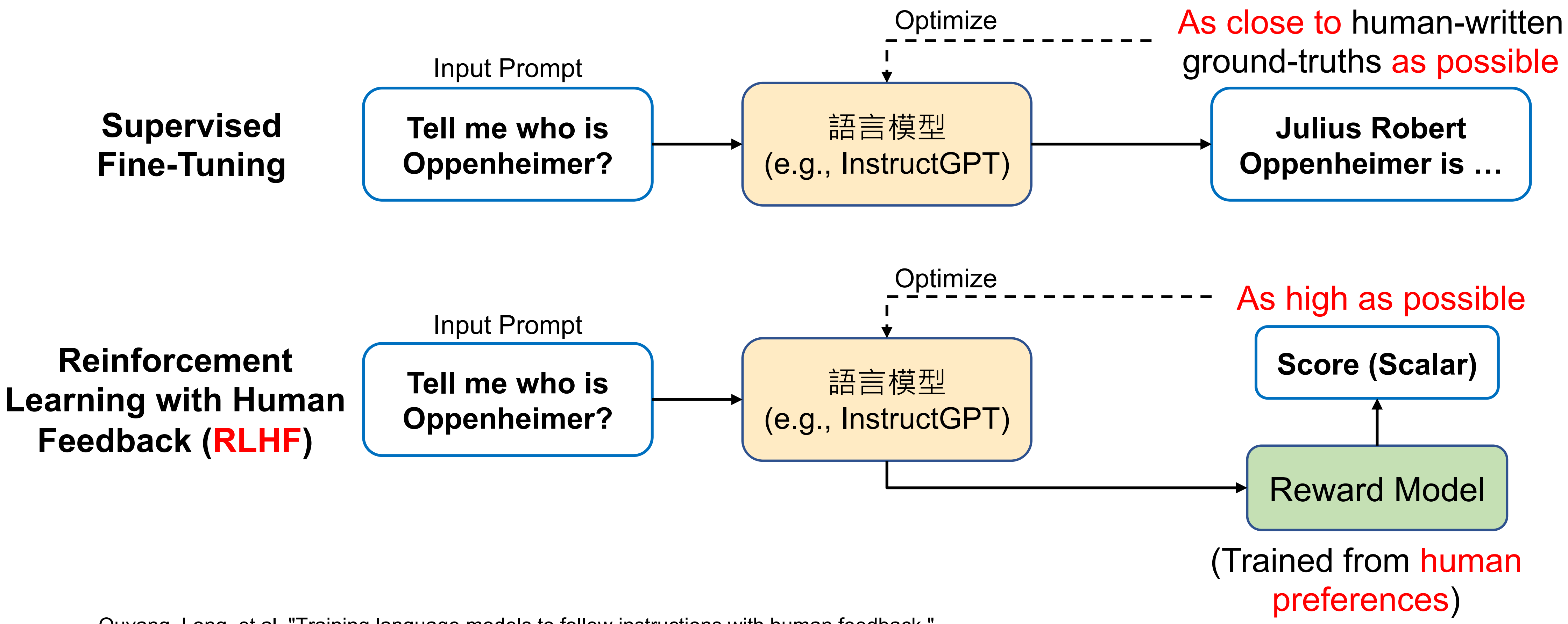
GPT-3 → **文字接龍 (Pre-training)** → **指令微調 (Supervised Fine-tuning, SFT)** → **人類反饋學習 (RLHF)** → InstructGPT

指令微調：讓模型**學會回答**

人類反饋學習：讓模型的回答貼近**人類想要**的回答

RLHF: Reinforcement Learning with Human Feedback

RLHF 的進行過程



Ouyang, Long, et al. "Training language models to follow instructions with human feedback." *Advances in neural information processing systems* 35 (2022): 27730-27744.

Main Differences between BERT and InstructGPT

Model size:

- BERT: 110 million
- InstructGPT: 175 billion (~=BERT*1590 times)

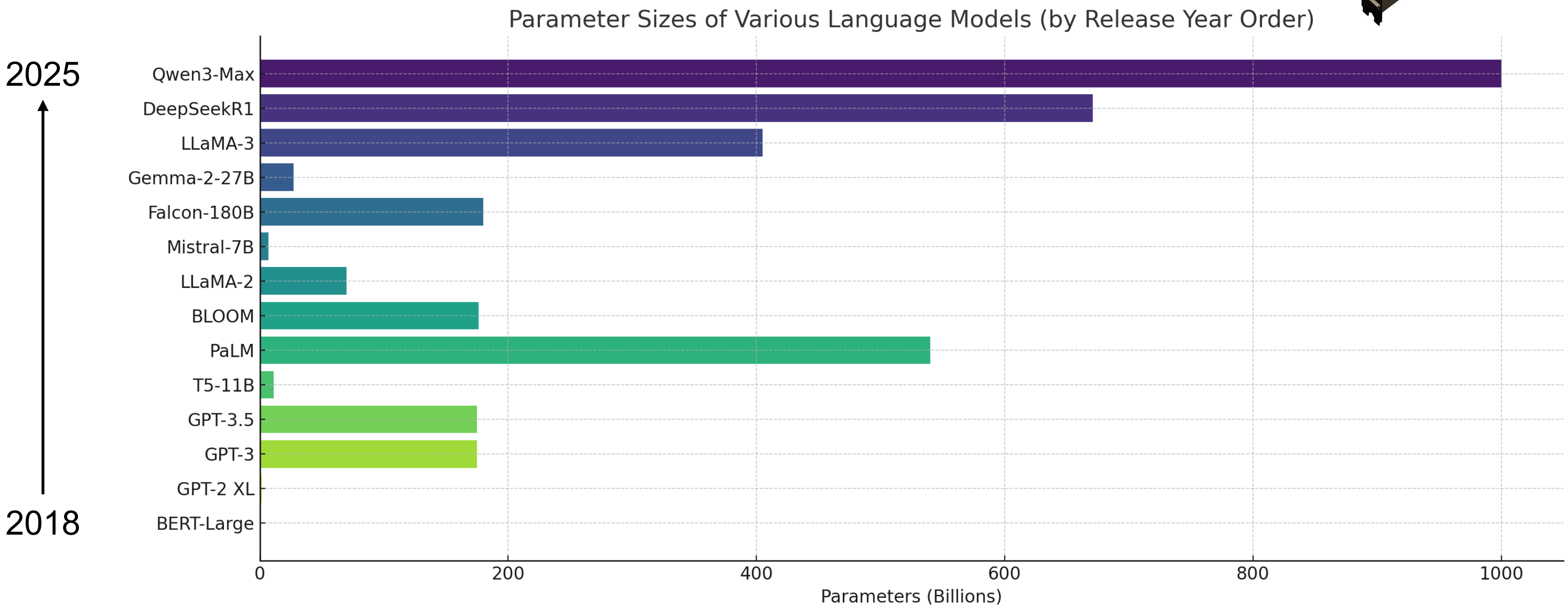
Training:

- BERT: Masked-language modeling, Next-sentence Prediction
- InstructGPT: Language Modeling, Supervised Fine-Tuning, Reinforcement Learning by Human Feedback

NLP Tasks:

- BERT: Natural Language Understanding (E.g., Classification)
- InstructGPT: Natural Language Generation (E.g., Machine Translation, Story Generation)

NLP Model Sizes (2018 – 2025)



The Revolution of ChatGPT

ChatGPT came out in November, 2022.

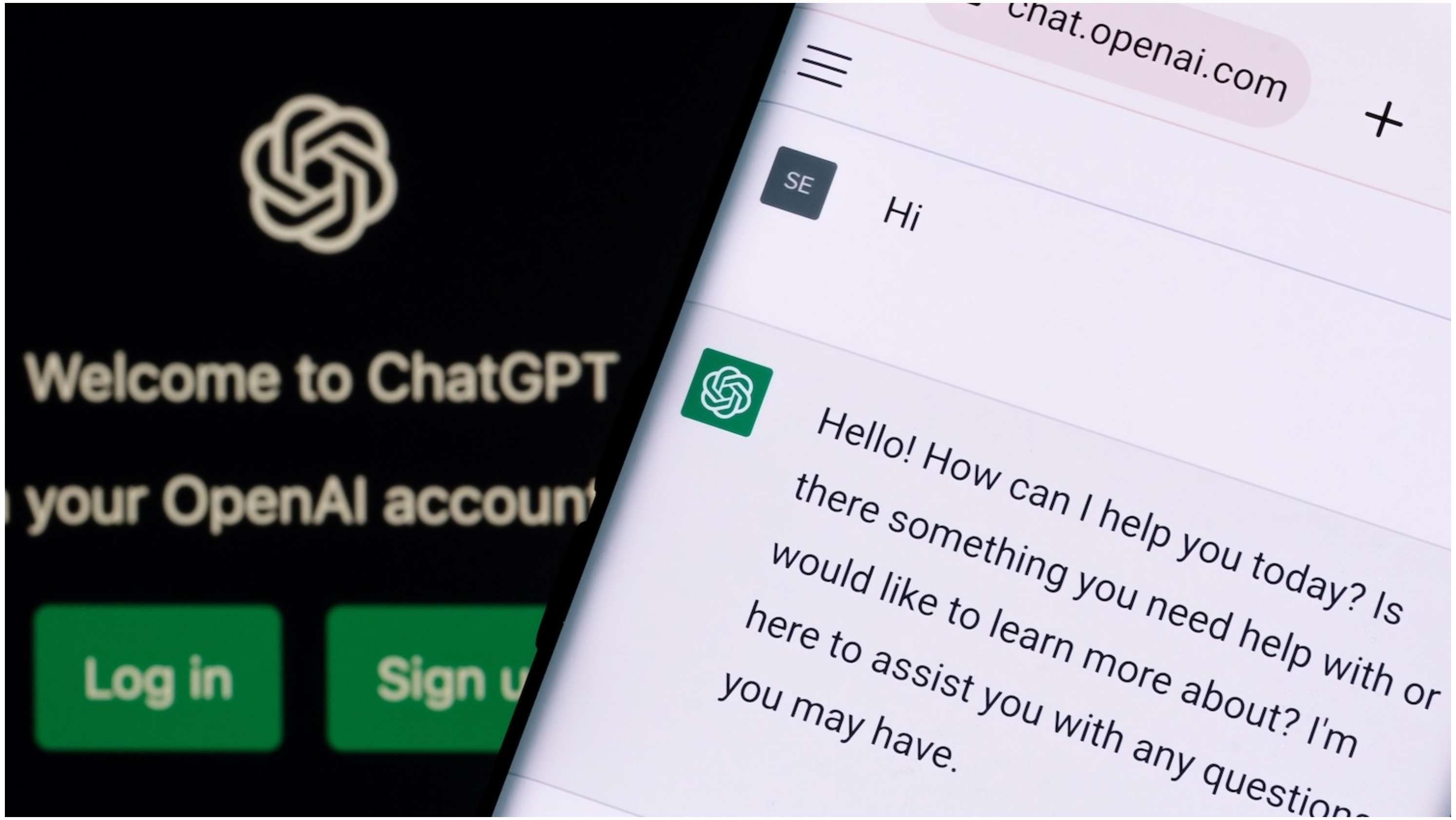
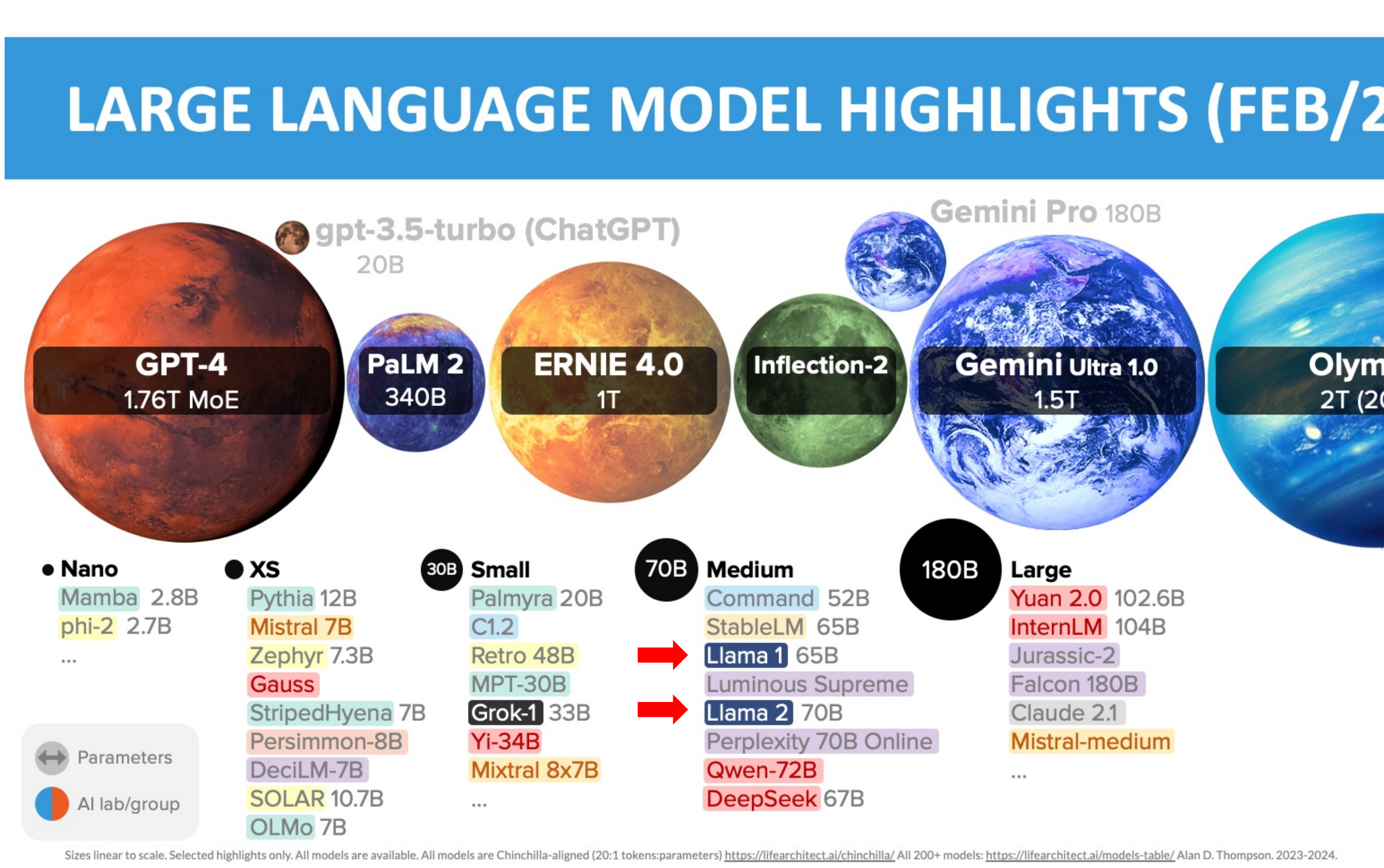


Figure source: https://poole.ncsu.edu/thought-leadership/article/lets-chat-about-chatgpt/
https://openai.com/index/chatgpt/

LLM Size Highlights



Source: Inside language models (from GPT-4 to PaLM) – Dr Alan D. Thompson – Life Architect

Source:https://commons.wikimedia.org/wiki/File:Llama_lying_down.jpg

很多厲害的LLMs都需要上網且不公開...

Meta

Llama (7B - 65B) by Meta (2023)

- Llama1: https://arxiv.org/abs/2302.13971
- Llama2: https://arxiv.org/abs/2307.09288
- Llama3:
- https://arxiv.org/abs/2407.21783

大型語言模型會思考？

NLP

Chain-of-Thought (CoT)

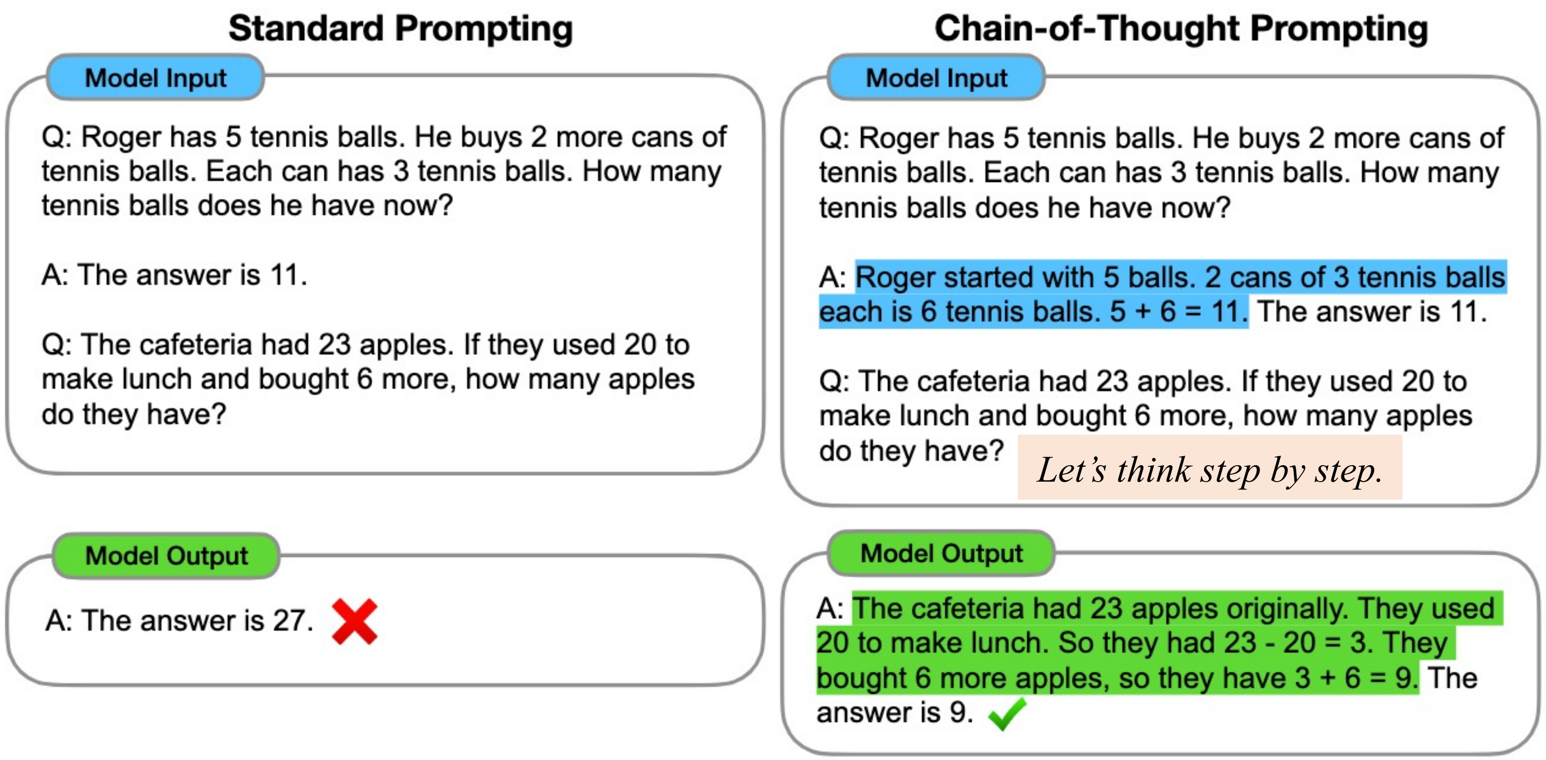


Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

CoT: https://arxiv.org/abs/2201.11903
Zero-shot-CoT: https://arxiv.org/abs/2205.11916

Training with Reasoning Tasks

The denominator of a fraction is 7 less than 3 times the numerator. If the fraction is equivalent to $2/5$, what is the numerator of the fraction? (Answer: $\boxed{14}$)

Let's call the numerator x.

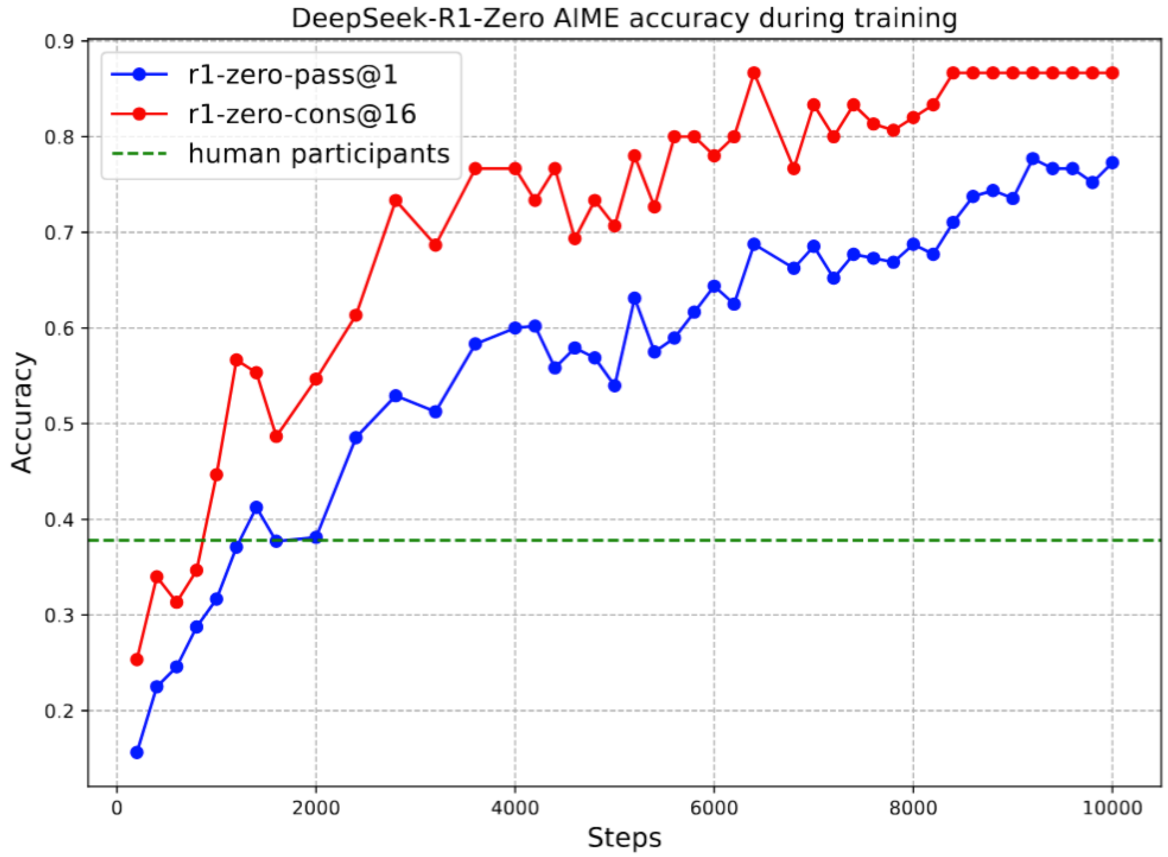
So the denominator is 3x-7.

We know that x/(3x-7) = 2/5.

So 5x = 2(3x-7).

5x = 6x - 14.

So x = 7.



Let's Verify Step by Step:
https://arxiv.org/abs/2305.20050
DeepSeek-R1:
https://arxiv.org/abs/2501.12948

我們這學期要學什麼？

這門課的目標 🎯

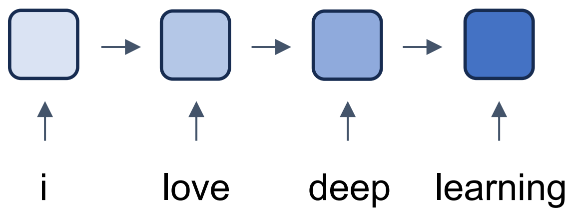
- 能夠「理解」主流 NLP / LLM 方法在做什麼
  - 不是只會背名詞、不是只會看別人的部落格 / 重點整理
- 能夠運用現成工具（尤其是 🤗 Transformers 生態）把模型跑起來、改得動
- 知道如何進階 ➡️ 評估、幻覺、安全性、RAG

- 如果你曾經覺得「LLM 很神，但我不知道它到底在神什麼」

—— 那這門課就是你的解謎本 📚🧩

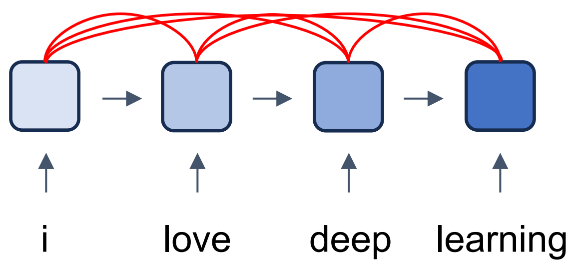
1. 自然語言處理基本功 (如何表達字詞)

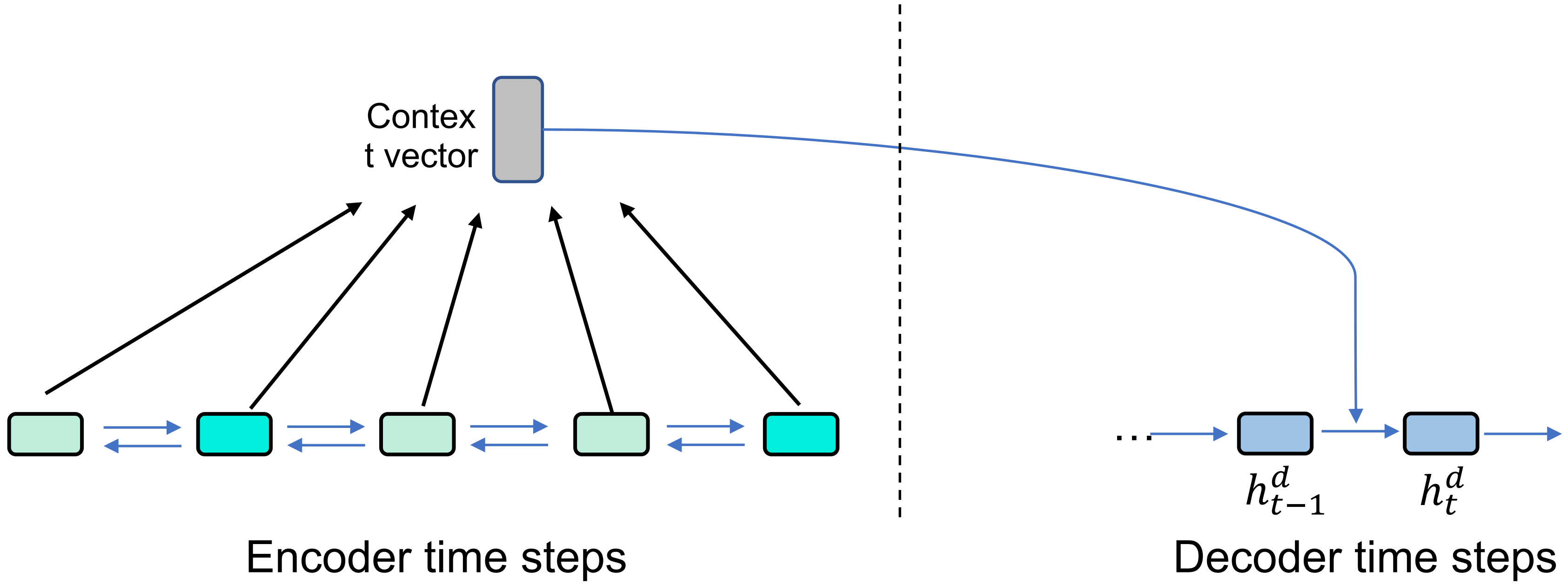
- Week 1 – Week 3
  - 自然語言處理介紹
  - 文字向量基礎與詞嵌入模型
  - 遞迴神經網路與長短期記憶網絡



2. 現代自然語言處理基本功

- Week 4 – Week 5
  - 注意力機制以及 Sub-word 分詞法
  - Transformer 架構



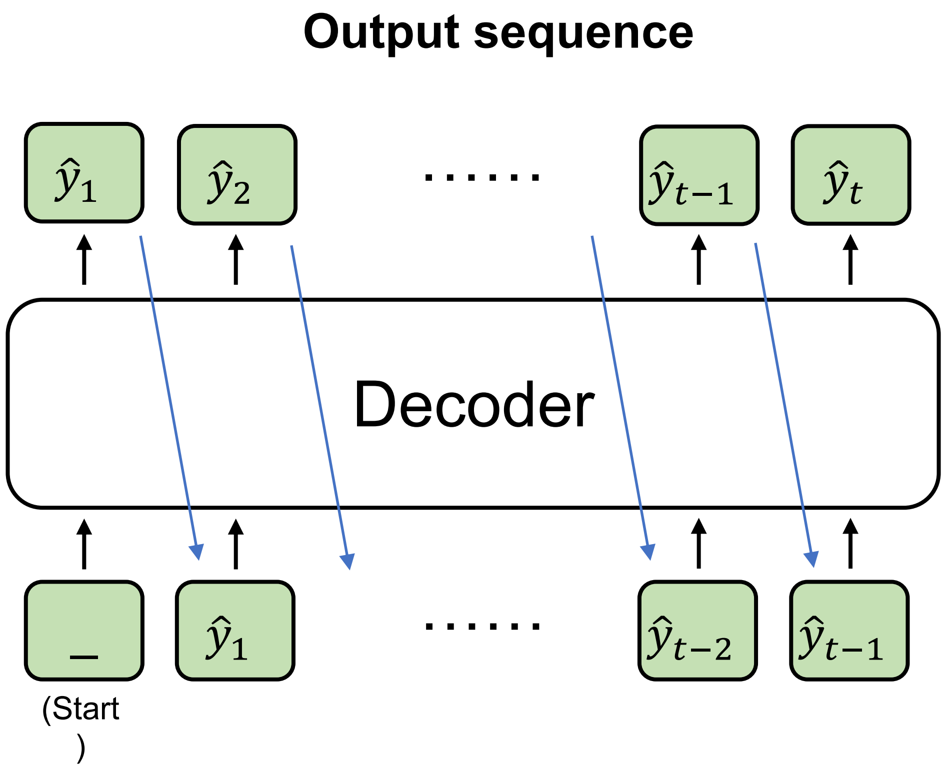


NLP

進階自然語言處理 (1)

OpenAI

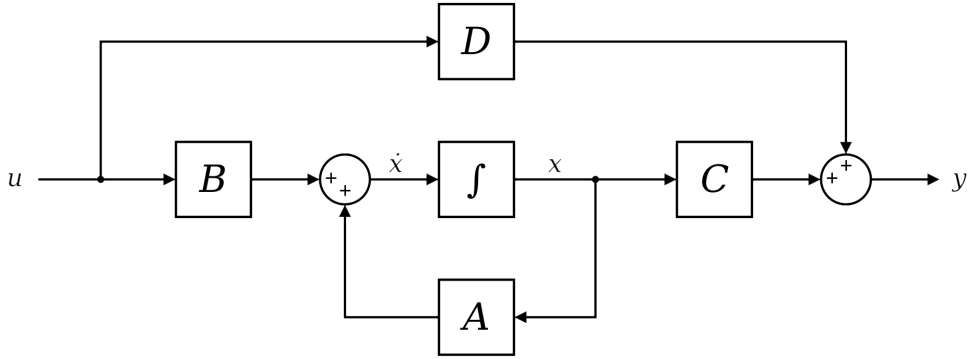
- Week 6 – Week 11
  - BERT  、GPT、文字生成策略
  - HuggingFace 函式庫實作教學
  - InstructGPT, RLHF, LLM
    - PPO, GRPO
  - Retrieval-augmented Generation



自然語言處理應用

deepseek

- Week 12 – Week 14
  - NLP Evaluations
    - Hallucinations? Safety?
  - 如何省力訓練大模型？➡️ Test-time Compute
  - State Space Models and Mamba



⬅️ Figure source:
https://huggingface.co/blog/lbourdois/get-on-the-ssm-train

Common NLP tasks

Text Classification	Text Generation	Sequence Tagging	Text Matching
Sentiment Analysis	Machine Translation	Part-of-speech tagging	Semantic Textual Similarity
Natural Language Inference (NLI)	Summarization	Named Entity Recognition (NER)	Retrieval
	Dialogue Generation	Word Segmentation	

我可以修這門課嗎？

- 建議有以下基礎：
  - ✅ 基礎 Deep Learning
  - ✅ 基礎 Python
  - ✅ 一點點機率/線代概念
- 願意花時間學習知識、學習除錯、學習驗證

Syllabus

Tutorial

Exam

Report

Week	Topic	Note
1	自然語言處理介紹以及本課程綱要	
2	文字向量基礎與詞嵌入模型	HW1
3	遞迴神經網路與長短期記憶網絡	
4	注意力機制與現代字詞分割演算法	
5	Transformer 模型架構介紹	HW2 (HW1 due)
6	BERT 與它的好夥伴：基於編碼器架構的 Transformer 模型	
7	清明連假	
8	GPT-1 到 GPT-3：基於解碼器架構的 Transformer 模型 文字生成策略	HW3 (HW2 due)
9	期中考	
10	HuggingFace 函式庫實作教學	
11	大型語言模型專題 1：指令微調與基於人類回饋的強化學習 & Retrieval Augmented Generation (RAG)	HW4 (HW3 due)
12	大型語言模型專題 2：如何評量模型的成效 / 幻覺 / 安全性？	
13	大型語言模型專題 3：如何訓練大模型？輕量化微調方法	
14	State Space Models and Mamba	(HW4 due)
15	小組實作成果報告 (1)	
16	小組實作成果報告 (2)	

PyTorch Tutorial

- PyTorch Basics:
  - https://youtu.be/nJpkmWbteTU?si=Ay2EEiXIWHDL9D-K
- PyTorch Modeling:
  - https://youtu.be/NEBJqya2IDs?si=-xX7dC_T3mDSAqjf
  - https://youtu.be/41r4nmeGitk?si=veztiILno9PUZuex

Scoring

- Assignments (4 times): 30%
- Quizzes: 1% x 10 times
- Mid-term Exam: 30%
- Term Project: 30%

About the Quizzes

- 不定時，代表出席分數，若多於10次即僅採計10次
- 與當週的上課內容有關
- Feel free to search the Internet

About the Assignments

- 與當週或鄰近週次的上課內容有關
- 作業一定都會有範例程式碼 (Sample Code)
- **需要寫 report**

作業可不可以用AI？

- 可以使用，但不得直接複製貼上
- 若使用 AI 生成程式碼，須以註解形式說明，例如：

```
class MLP(nn.Module):
    def __init__(self):
        # 以下幾行由gpt-5生成
        # 建構MLP的輸入層、中間層、以及輸出層
        self.layer_1 = nn.Linear(28*28, hidden_size1)
        self.layer_2 = nn.Linear(256, hidden_size2)
        self.cls_layer = nn.Linear(hidden_size2, 10)
        self.relu = nn.ReLU()
```

Final Project 是什麼？

- Final Project 是一種完整的訓練
- 你將自行從頭到尾做完一個題目，包含：
  - 資料前處理 (Data Pre-processing)
  - 模型訓練 (Model Training)
  - 模型評估 (Evaluations) -->了解問題並改進模型
- 每組人數待修課同學總數訂定，原則上最多 4 人/組

Textbook

Speech and Language Processing (3rd ed. draft)

Dan Jurafsky and **James H. Martin**

Here's our January 12, 2025 release!

This release has no new chapters, but fixes typos and also adds new slides and updated old slides. Individual chapters and updated slides are below.

Here is a single pdf of Jan 12, 2025 book!

1. **Feel free to use the draft chapters and slides in your classes**, print it out, whatever, the resulting feedback we get from you makes the book better!
2. **Typos and comments** are very welcome (just email slp3edbugs@gmail.com and let us know the date on the draft)! (Don't bother reporting missing refs due to cross-chapter cross-reference problems in the indvidual chapter pdfs, those are fixed in the full book draft)
3. **Gratitude!** We've put up a list here of the amazing people who have sent so many fantastic suggestions and bug-fixes for improving the book. We are really grateful to all of you for your help, the book would not be possible without you!
4. **How to cite the book**:

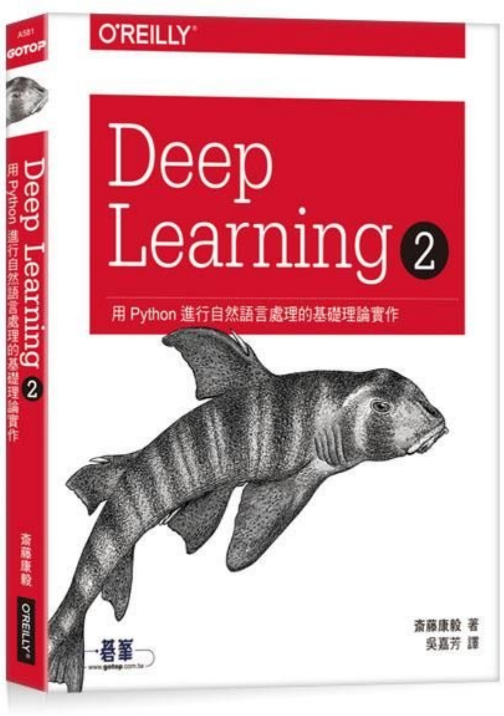
   Daniel Jurafsky and James H. Martin. 2025. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models, 3rd edition. Online manuscript released January 12, 2025. https://web.stanford.edu/~jurafsky/slp3.

5. A **bib entry** for the book is **here**.
6. **When** will the book be finished? Don't ask.
7. If you need the previous Aug 2024 draft chapters, they are here;

Speech and Language Processing (3rd ed. draft) Dan Jurafsky and James H. Martin.

Free Online book (Jan 2026 version): https://web.stanford.edu/~jurafsky/slp3/

Reference book



Deep Learning 2 | 用Python進行自然語言處理的基礎理論實作，歐萊禮 (ISBN: 9789865020675)

GitHub of this book:
https://github.com/oreilly-japan/deep-learning-from-scratch-2

生成式AI融合教學

- 以組為單位
  - ~~ChatGPT Plus~~ Claude Pro
  - Google Colab Pro 雲端運算付費服務
  - OpenAI /Gemini API 呼叫大型語言模型進行生成式服務

Contact Information

Instructor: 林英嘉

✉️ yjlin@cgu.edu.tw

🕘 Office Hour: 歡迎來信預約，一定沒問題

寄信前請在主旨加註記 [自然語言處理]

Thank you!

Instructor: 林英嘉

✉️ yjlin@cgu.edu.tw

TA: 林宣辰

✉️ b1128015@cgu.edu.tw